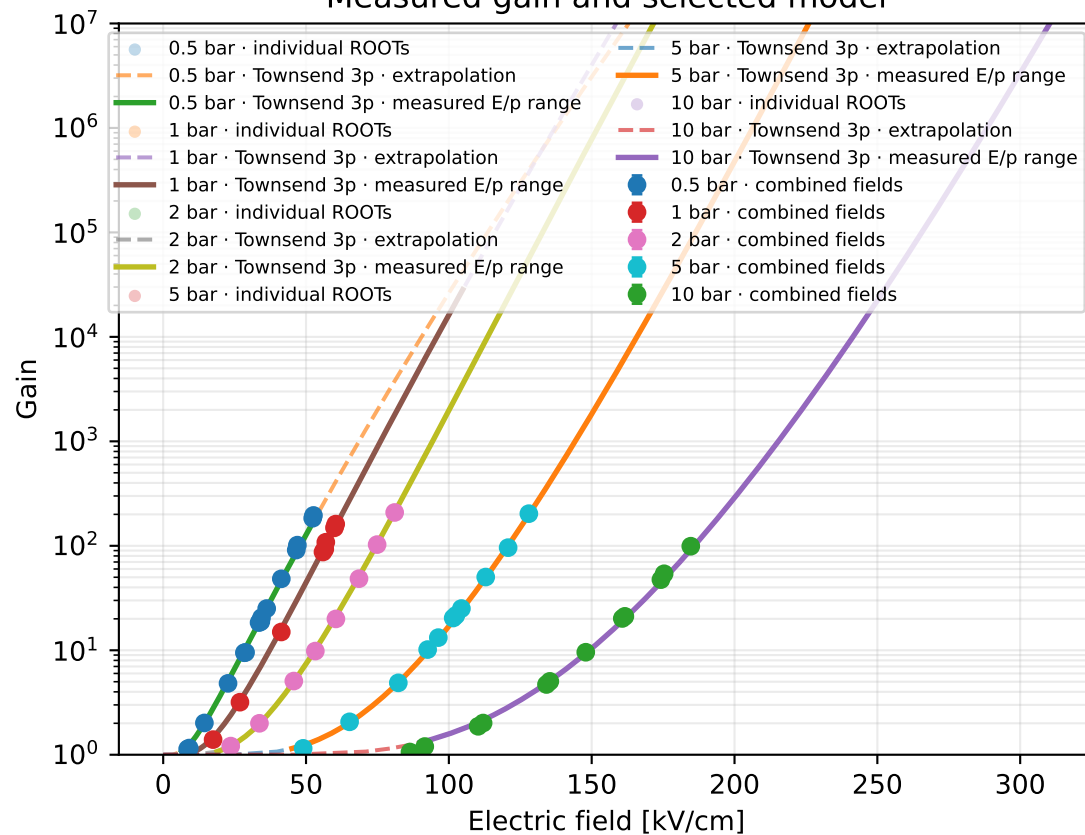
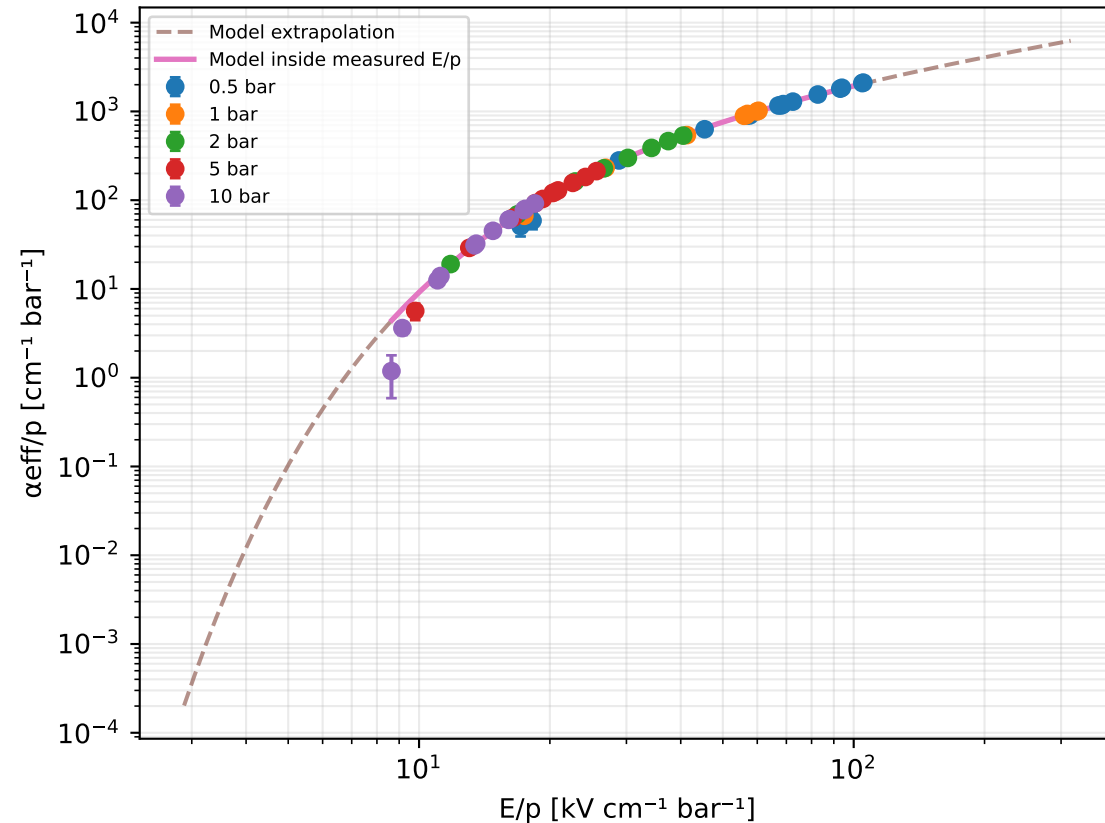


ArCF4 · Ar 99 % / CF4 1 % · gap 0.05 mm

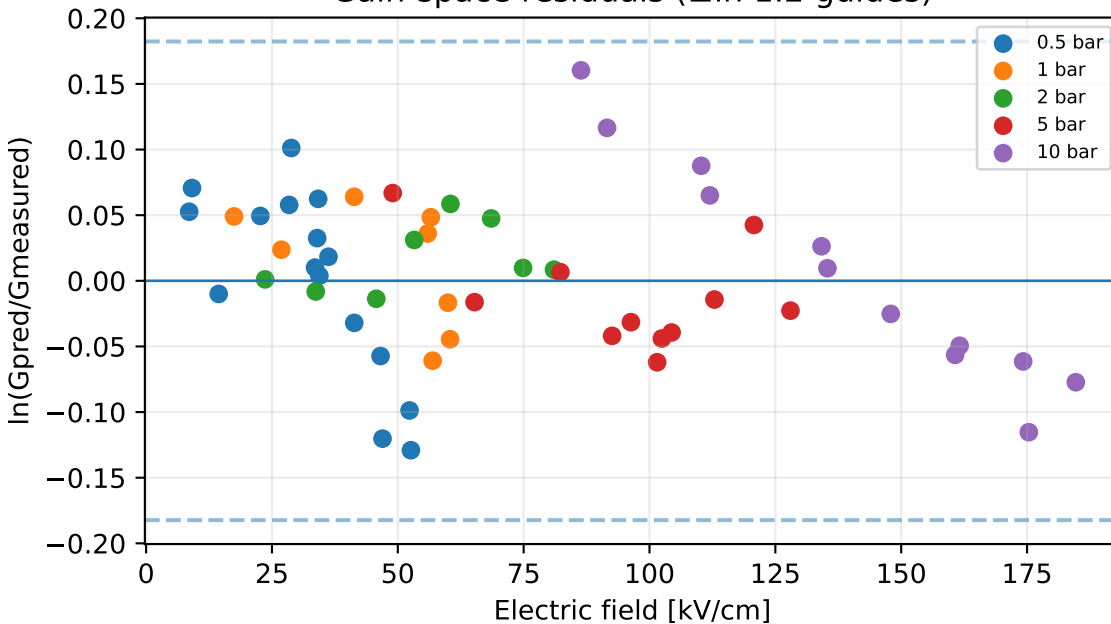
Measured gain and selected model



Reduced Townsend fit

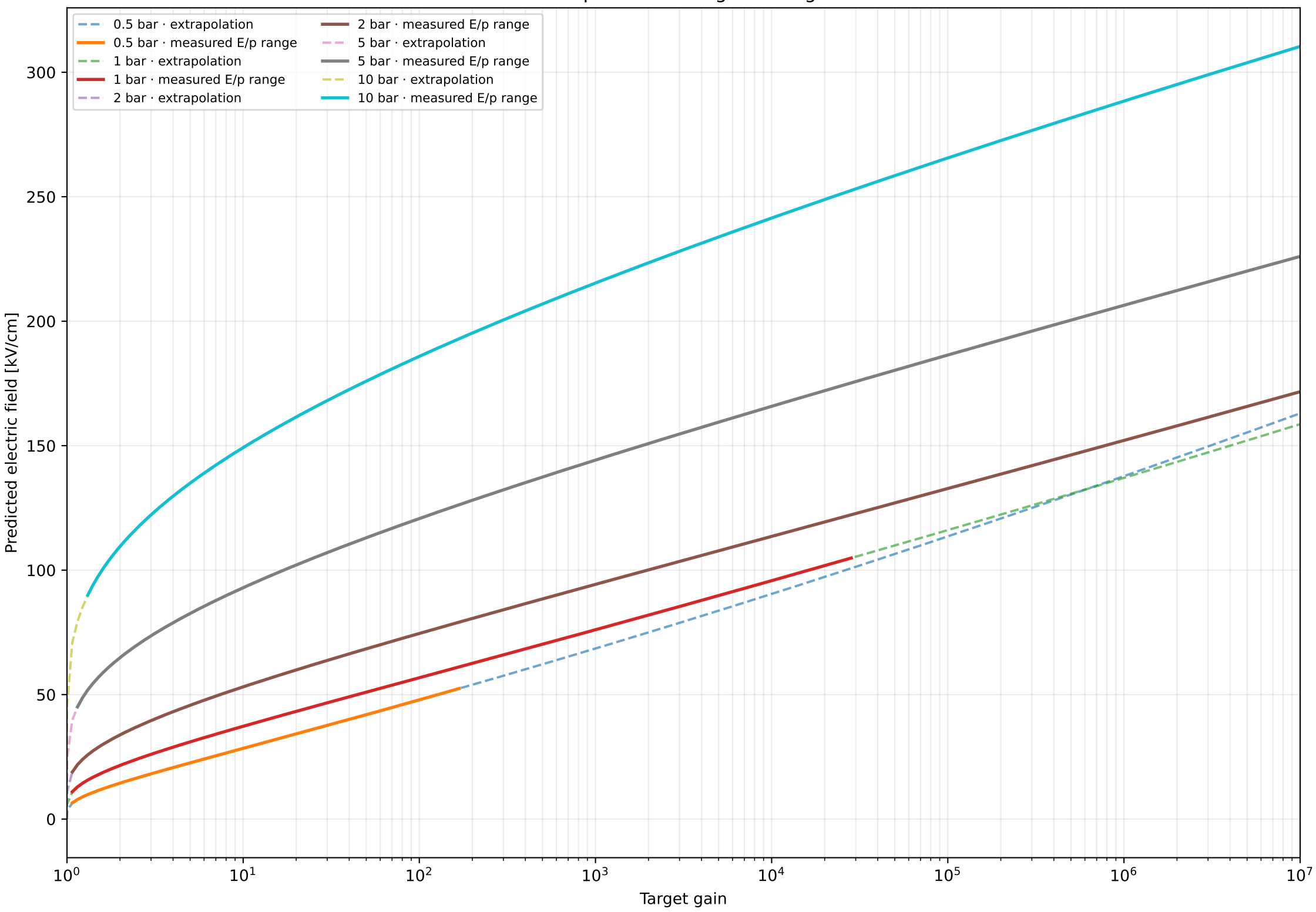


Gain-space residuals ($\pm \ln 1.2$ guides)



ROOT runs: 75
 Unique physical points: 55
 Pressures: 0.5, 1, 2, 5, 10 bar
 Selected model: Townsend 3p
 Status: accepted
 Training RMSE: $\times 1.234$
 Cross-validation median: $\times 1.050$
 Cross-validation worst: $\times 1.175$
 Reduced-field range: 8.639–105.2 kV cm $^{-1}$ bar $^{-1}$
 A=77.215, B=39.396, m=0.78553, n=1
 Covariance condition: 1.6e+04

Field extrapolation from gain 1 to gain 10^7



p [bar]	E [kV/cm]	E/p	Gain	σGain	αeff [cm ⁻¹]	npe	runs
0.5	8.5613	17.1226	1.135922	0.0339	25.489	103	1
0.5	9.1222	18.2444	1.15873	0.0346	29.4649	126	1
0.5	14.4223	28.8447	2.015504	0.0605	140.174	387	1
0.5	22.6973	45.3947	4.824561	0.145	314.744	456	1
0.5	28.4152	56.8305	9.469734	0.284	449.62	413	1
0.5	28.8371	57.6742	9.538941	0.286	451.076	321	1
0.5	33.5631	67.1261	18.41032	0.552	582.582	407	1
0.5	33.9846	67.9691	18.93103	0.567	588.161	435	1
0.5	34.1809	68.3617	18.8093	0.564	586.87	430	1
0.5	34.4222	68.8444	20.52647	0.614	604.343	340	1
0.5	36.2317	72.4634	25.09948	0.752	644.569	382	1
0.5	41.3467	82.6934	48.30925	1.45	775.525	346	1
0.5	46.5748	93.1495	91.09288	2.73	902.376	323	1
0.5	46.9649	93.9298	101.4815	3.04	923.975	324	1
0.5	52.3353	104.671	183.7	5.52	1042.66	300	1
0.5	52.6061	105.212	195.2609	5.85	1054.87	299	1
1	17.4658	17.4658	1.39604	0.0419	66.7279	303	1
1	26.8704	26.8704	3.18481	0.0955	231.679	395	1
1	41.3391	41.3391	14.96917	0.448	541.199	519	1
1	55.934	55.934	87.20667	3.15	893.656	300	1
1	56.5628	56.5628	92.88076	2.78	906.263	369	1
1	56.9254	56.9254	108.2156	3.24	936.825	473	1
1	59.9565	59.9565	148.9659	1.24	1000.74	5697	13
1	60.3914	60.3914	161.3733	5.8	1016.74	300	1
2	23.6385	11.8193	1.21	0.0362	38.1241	300	1
2	33.6966	16.8483	2	0.0599	138.629	529	1
2	45.7242	22.8621	5.073899	0.152	324.822	636	1
2	53.2562	26.6281	9.827941	0.295	457.046	680	1
2	60.4596	30.2298	19.96154	0.599	598.761	650	1
2	68.5436	34.2718	48.46128	1.45	776.153	594	1

p [bar]	E [kV/cm]	E/p	Gain	σ Gain	α_{eff} [cm ⁻¹]	npe	runs
2	74.886	37.443	102.7038	3.08	926.37	584	1
2	81.0317	40.5158	208.53	8.49	1068.02	300	1
5	48.9938	9.79876	1.151613	0.0345	28.2327	310	1
5	65.252	13.0504	2.066274	0.062	145.149	679	1
5	82.3055	16.4611	4.878716	0.146	316.976	841	1
5	92.5742	18.5148	10.15666	0.305	463.626	766	1
5	96.3099	19.262	13.21798	0.396	516.316	890	1
5	101.517	20.3033	20.36302	0.611	602.744	887	1
5	102.457	20.4915	21.55388	0.647	614.111	863	1
5	104.395	20.8791	25.08919	0.752	644.487	740	1
5	112.905	22.5811	50.23899	1.51	783.358	749	1
5	120.743	24.1486	96.01667	3.54	912.904	600	1
5	128.003	25.6007	203.0967	9.09	1062.74	300	1
10	86.3929	8.63929	1.061189	0.0318	11.878	572	1
10	91.5661	9.15661	1.198488	0.0359	36.2121	529	1
10	110.271	11.0271	1.871622	0.0561	125.361	888	1
10	111.991	11.1991	2.011527	0.0603	139.779	694	1
10	134.173	13.4173	4.705515	0.141	309.747	961	1
10	135.348	13.5348	5.039153	0.151	323.448	945	1
10	147.927	14.7927	9.578799	0.287	451.91	1066	1
10	160.714	16.0714	20.1654	0.605	600.794	919	1
10	161.613	16.1613	21.12755	0.633	610.116	933	1
10	174.232	17.4232	47.45819	1.42	771.97	873	1
10	175.355	17.5355	53.98642	1.62	797.746	957	1
10	184.686	18.4686	99.14209	0.993	919.311	8361	9

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	-188	×1.22	×4.36	no	large cross validation outlier
Townsend 3p	3	-165	×1.05	×1.17	yes	—
Townsend 4p	4	-191	×1.03	×1.14	yes	—